LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 13 Test: Our Environment

Chapter / Topic: Our Environment

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.

SECTION A

6 marks

Choose the correct option.

Q1. Which group contains only biodegradable items? **[1]**

(a) Grass, wood and plastic
(b) Fruit peels, cake and lime juice
(c) Glass, paper and leaves
(d) Polythene, cotton and wool

Q2. Which of the following forms a food chain? **[1]**

(a) Grass, wheat and mango (b) Goat, cow and elephant
(c) Grass, goat and human (d) Grass, fish and goat

Q3. Which of the following are environment-friendly practices? **[1]**

(a) Carrying cloth bags for shopping
(b) Switching off unnecessary lights and fans
(c) Walking to school instead of being dropped by scooter
(d) All of the above

Q4. In a food chain, the energy passed on to the next trophic level is about: **[1]**

(a) 10% of the previous level (b) 50% of the previous level
(c) 90% of the previous level (d) the same as the previous level

Q5. Depletion of the ozone layer is mainly caused by: **[1]**

(a) carbon dioxide (b) chlorofluorocarbons (CFCs)
(c) oxygen (d) nitrogen

Q6. **Assertion (A):** Food chains generally have only three or four trophic levels. **Reason (R):** The energy available decreases greatly at each successive trophic level. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. What are decomposers? What would happen if there were no decomposers? **[2]**

OR

How is ozone formed in the upper atmosphere? Write the equations.

Q8. What is biological magnification? Why is it highest in organisms at the top of a food chain? **[2]**

SECTION C

6 marks

Q9. Study the food chain: Grass → Grasshopper → Frog → Snake → Eagle **[3]**

(i) Identify the producer and the tertiary consumer. **[1]**

(ii) If 10,000 J of energy is available at the producer level, how much energy will reach the snake? **[1]**

(iii) Which organism will have the highest concentration of a pesticide that enters this food chain? Why? **[1]**

Q10. Why is damage to the ozone layer a cause for concern? What steps have been taken to limit it? **[3]**

OR

Suggest three ways in which you can reduce the waste your household produces.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) What is an ecosystem? Name its two types of components, with examples. **[2]**

(ii) Explain the flow of energy in an ecosystem. Why is it unidirectional? **[3]**

OR

(a) Differentiate between biodegradable and non-biodegradable substances, with two examples of each. [2]
(b) Give three ways in which non-biodegradable substances harm the environment. [3]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. Some years ago, it was proposed that tea on trains should be served in disposable kulhads (clay cups) instead of plastic cups. It was soon realised that making kulhads on such a large scale would cause the loss of fertile top soil. Now, disposable paper cups are commonly used. **[4]**

(i) Why was it proposed to replace plastic cups? **[1]**

(ii) What was the problem with using kulhads on a large scale? **[1]**

(iii) Give one advantage of paper cups over plastic cups and one over kulhads. **[2]**

~ All the best! ~